

SMART TRAFFIC SIGNAL CONTROLLER USING STM32 BLUE PILL

Procedure Report

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Submitted to

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Procedure (Installation to Output) Report

1. Project Overview

This project explains how to build a smart traffic signal system using the STM32 Blue Pill microcontroller. The system is designed for a **four-way intersection**, where traffic from four directions is controlled using LEDs. Each road has three signals: **Red, Yellow, and Green**. The system changes signals in a proper sequence to avoid collisions and ensure smooth traffic flow. FreeRTOS is used to manage timing and task execution. Push buttons are used to simulate traffic conditions and extend green signal time when needed. A TTL module (YP05) is included as a communication component.

This project helps beginners understand:

- Embedded systems
- GPIO control
- RTOS concepts

2. Objective

- To design a four-way traffic signal system
- To control signals using STM32
- To implement multitasking using FreeRTOS
- To simulate real-time traffic control
- To understand UART communication basics

3. Hardware Components

- STM32 Blue Pill (STM32F103C8T6)
- LEDs (Red, Yellow, Green – 4 sets → total 12 LEDs)
- Resistors (220Ω)
- Push Buttons (2 or more)
- Breadboard
- Connecting wires
- USB cable
- YP05 TTL Module

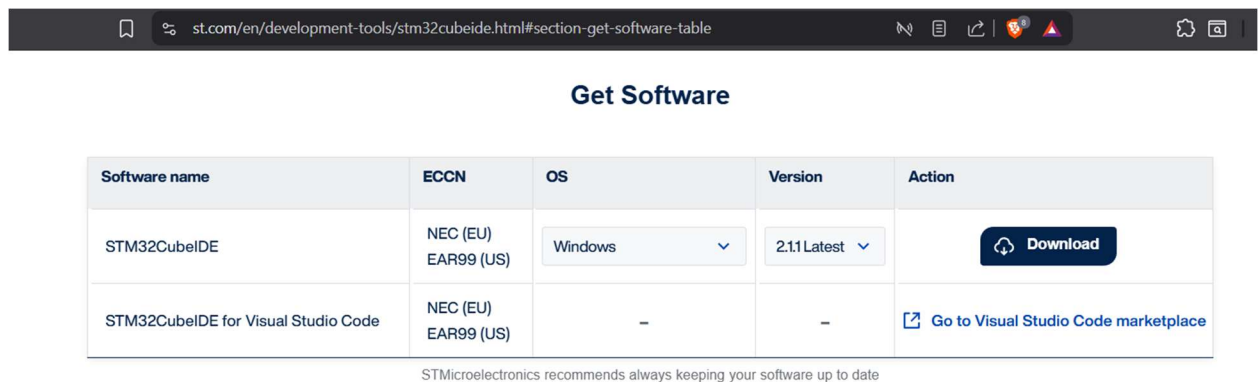
4. Software Requirements

- STM32CubeIDE
- FreeRTOS (inside CubeMX)
- STM32CubeProgrammer
- Serial Monitor (optional)

5. Software Installation

Step 1: Download STM32CubeIDE from:

<https://www.st.com/en/development-tools/stm32cubeide.html>



Get Software

Software name	ECCN	OS	Version	Action
STM32CubeIDE	NEC (EU) EAR99 (US)	Windows	2.11.Latest	Download
STM32CubeIDE for Visual Studio Code	NEC (EU) EAR99 (US)	–	–	Go to Visual Studio Code marketplace

STMicroelectronics recommends always keeping your software up to date

Fig 1. Installation of STM32CubeIDE

Step 2 :Download STM32CubeProgrammer:

<https://www.st.com/en/development-tools/stm32cubeprog.html>

Part Number	General Description	Latest version	Supplier	Download	All versions
+ CubePrg-Mac-Arm	STM32CubeProgrammer software for MacOS Arm	2.22.0	ST	Get latest	Select version ▼
+ STM32CubePrg-Lin	STM32CubeProgrammer software for Linux	2.22.0	ST	Get latest	Select version ▼
+ STM32CubePrg-Mac	STM32CubeProgrammer software for Mac	2.22.0	ST	Get latest	Select version ▼
+ STM32CubePrg-W32	STM32CubeProgrammer software for Win32	2.22.0	ST	Get latest	Select version ▼
+ STM32CubePrg-W64	STM32CubeProgrammer software for Win64	2.22.0	ST	Get latest	Select version ▼

STMicroelectronics recommends always keeping your software up to date

Fig 2. Installation of STM32 CIBE PROGRAMMER

Select the latest version win32/win64 according to your Pc

6. Hardware Connections

LED Connections (4 Roads)

Road 1:

- PA0 → Red
- PA1 → Yellow
- PA2 → Green

Road 2:

- PA3 → Red
- PA4 → Yellow
- PA5 → Green

Road 3:

- PA6 → Red
- PA7 → Yellow

- PB0 → Green

Road 4:

- PB1 → Red
- PB10 → Yellow
- PB11 → Green

Button Connections

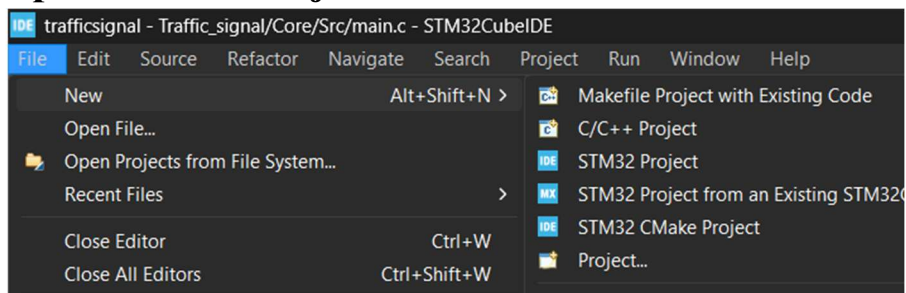
- PB12 → Button 1
- PB13 → Button 2

YP05 TTL Module :

- VCC → 3.3V
- GND → GND
- TX → PA10
- RX → PA9

7. Software Implementation

Step 1: Create Project



- Open STM32CubeIDE
- Click “New STM32 Project”

- Select STM32F103C8
- Click Finish

Step 2: Configure Pins

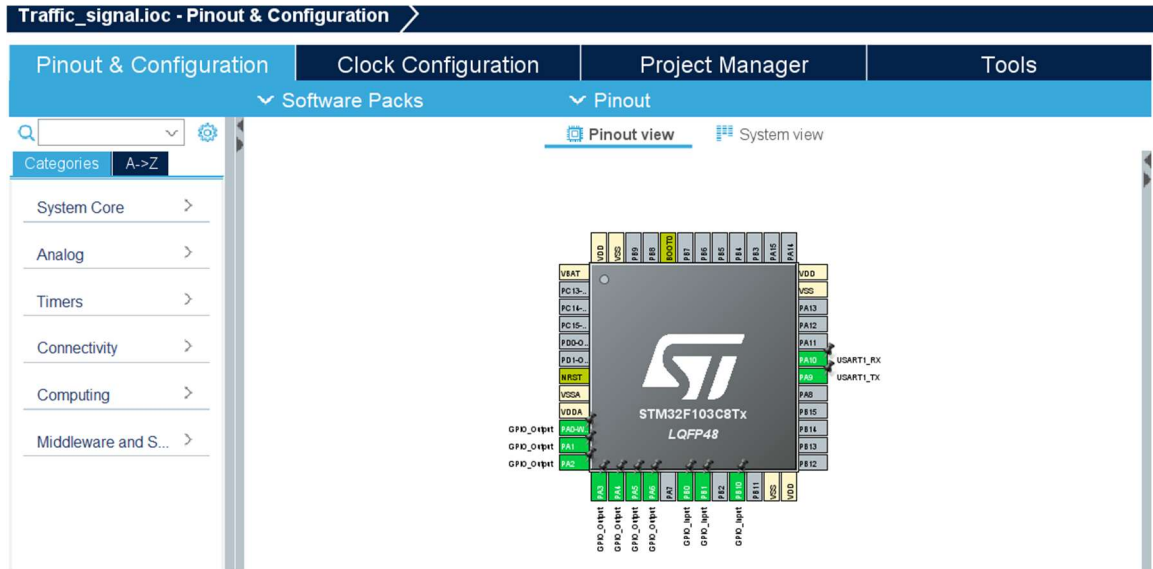


Fig 3. Configuring STM32 Pins

Click pins and set:

- PA0–PA7 → Output
- PB0, PB1, PB10, PB11 → Output
- Buttons → Input

Step 3: Enable FreeRTOS

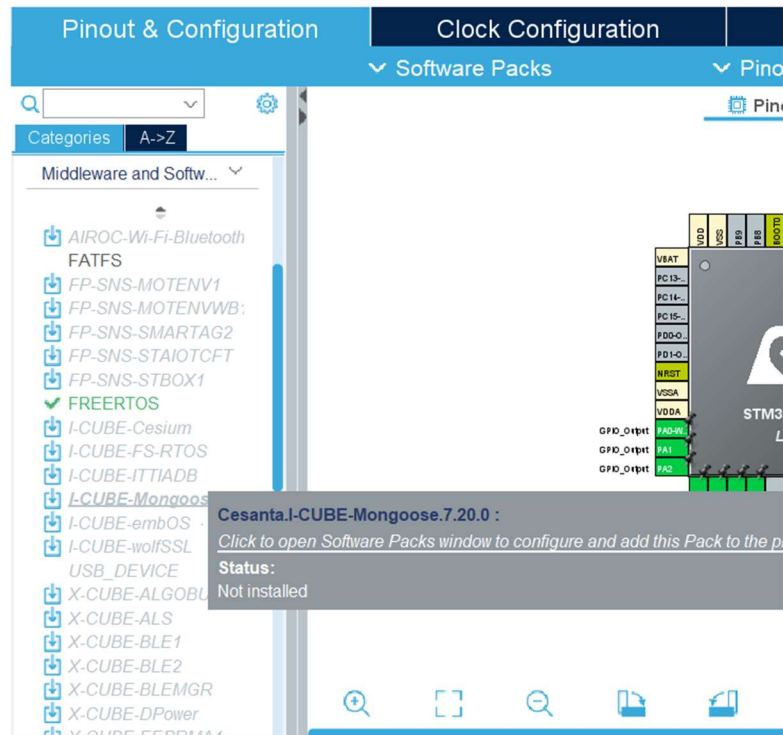


Fig 4. Configuring RTOS

- Go to Middleware
- Enable FreeRTOS

Step 4: Generate Code

- Click Generate Code

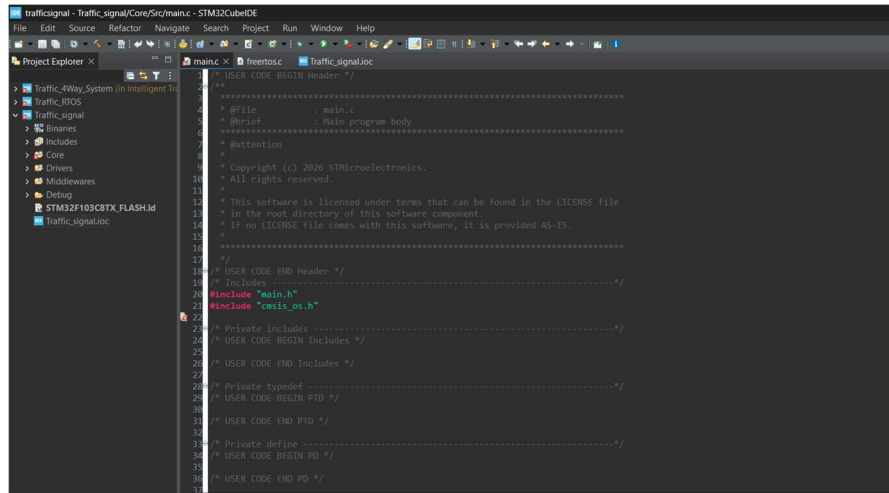


Fig 5. main.c file after generation

- You will get this kind of main.c file after generating choosing “Generate code” option

Step 5: Write Logic

Inside main.c write your logic according to our project (refer code report)for the logic

Step 6: Build Project

- Click Build
- It will generate .hex file

Step 7: Upload Code(using STM32 CUBE PROGRAMMER)

- Connect STM32
- Check COM port and connect it using TTL (YP05) and now we will have our hex file

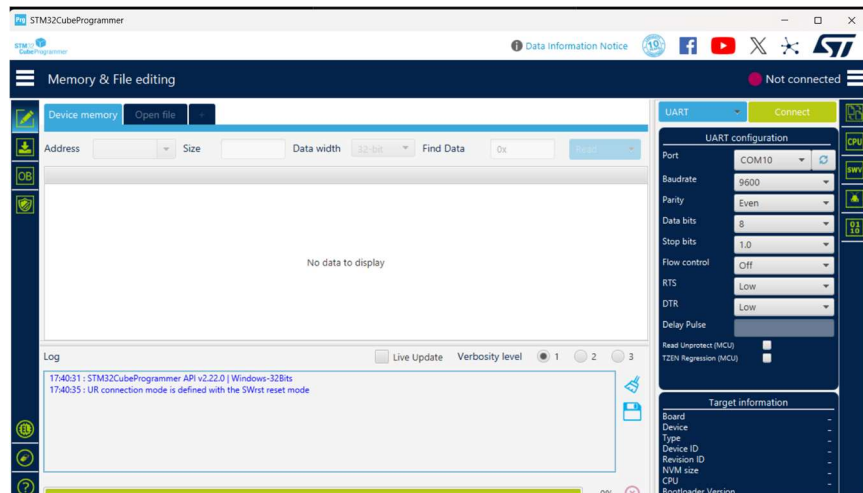


Fig 6.STM32 CUBE PROGRAMMER interface

- Select open file and choose .hex file which is generated and upload it to board.

8. Telegram Bot Set Up

Step 1: Create Bot

- Open Telegram
- Search **BotFather**
- Type /start
- Type /newbot
- Enter bot name
- Copy BOT TOKEN

Step 2: Get Chat ID

- Search **userinfobot**
- Type /start
- Copy Chat ID

Step 3: Python Code to Send Message

- Install requests
- Use Python script to send message

9. Excel Logging

Step 1: Install pandas

```
pip install pandas openpyxl
```

Step 2: Logging Data

- Store:
 - Time
 - Signal status
- Save in .xlsx file

10. Working

- Only one road gets green
- Others remain red
- Yellow used for transition
- Buttons extend time
- STM32 sends data via UART
- Python reads data
- Telegram sends updates
- Excel stores logs

11. Output

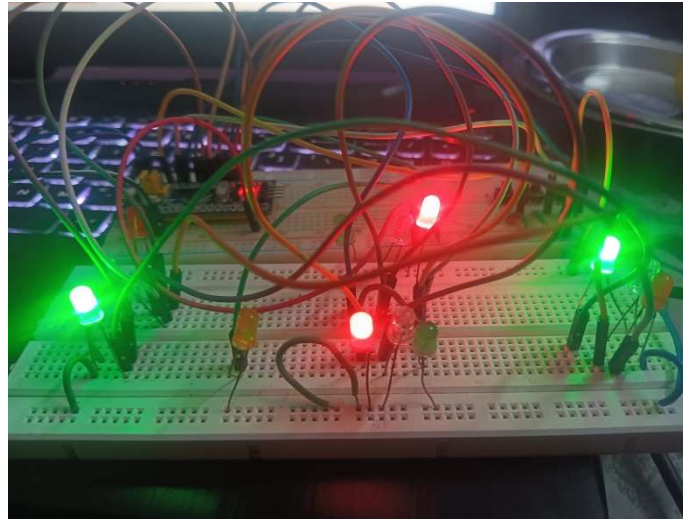


Fig 7.Setup and Output

12. Results

Home traffic_analysis.xlsx				
File Home Insert Page Layout Formulas Data Review View Tools Smart Toolbox				
Format Painter Paste Font Paragraph Styles Orientation Merge and Center				
A18 fx EW PRIORITY EXTENDED				
	A	B	C	D
1	State	Start	End	Duration
2	BOTH YELLOW	13:04:44	13:04:45	1
3	NS GREEN	13:04:45	13:04:47	1.47
4	BOTH YELLOW	13:04:47	13:04:49	2.19
5	EW GREEN	13:04:49	13:04:50	1.23
6	BOTH YELLOW	13:04:50	13:04:52	1.41
7	NS GREEN	13:04:52	13:04:54	2.23
8	BOTH YELLOW	13:04:54	13:04:55	1.14
9	EW GREEN	13:04:55	13:04:56	1.12
10	BOTH YELLOW	13:04:56	13:04:57	1.25
11	NS GREEN	13:04:57	13:04:59	1.65
12	BOTH YELLOW	13:04:59	13:05:00	1.22
13	EW GREEN	13:05:00	13:05:03	2.41
14	BOTH YELLOW	13:05:03	13:05:04	1.14
15	NS GREEN	13:05:04	13:05:05	1.23
16	BOTH YELLOW	13:05:05	13:05:06	1.2
17	EW GREEN	13:05:06	13:05:07	1.2
18	EW PRIORITY EXTENDED	13:05:07	13:05:13	5.63
19	BOTH YELLOW	13:05:13	13:05:14	1.2
20	NS GREEN	13:05:14	13:05:15	1.29
21	BOTH YELLOW	13:05:15	13:05:17	1.15
22	EW GREEN	13:05:17	13:05:18	1.22
23	BOTH YELLOW	13:05:18	13:05:21	3.65
24	NS GREEN	13:05:21	13:05:25	3.88
25	BOTH YELLOW	13:05:25	13:05:29	3.6

Fig 8.Excel Sheet Logged Images

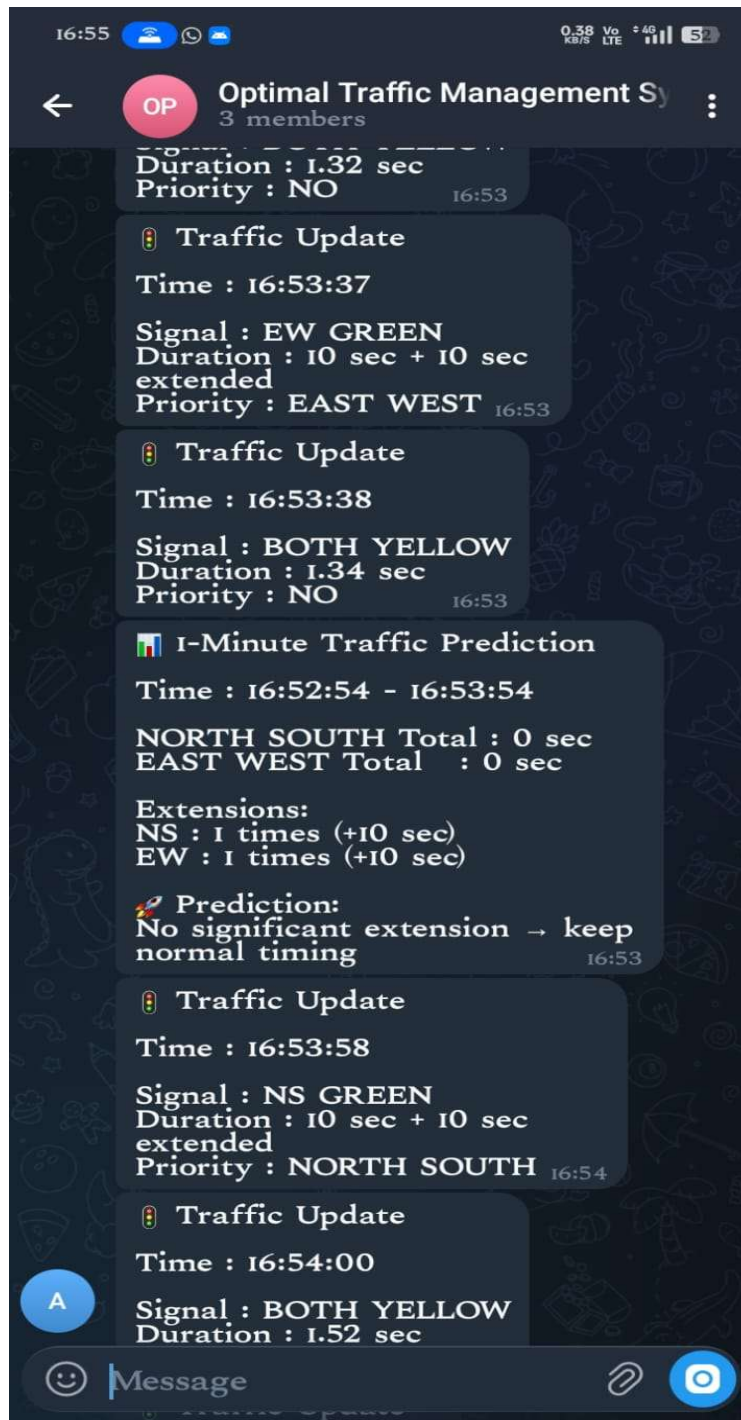


Fig 9. Telegram report images